
- +
 - -

LARGE LANGUAGE MODELS



? What exactly are
LLMs

? Why are they causing
such a stir

LLMs

A diagram with a central white circle containing the text "LLMs". Surrounding this central circle are eight teal-colored circles, each containing the name of a different Large Language Model. The circles are arranged in a roughly circular pattern around the center. The background is dark blue with a faint, glowing network of white lines and dots, suggesting a digital or neural network theme.

Alpaca

Llama

BLOOM

WizardLM

Dolly

Cohere

MPT

Bard



TRANSFORMER ARCHITECTURE

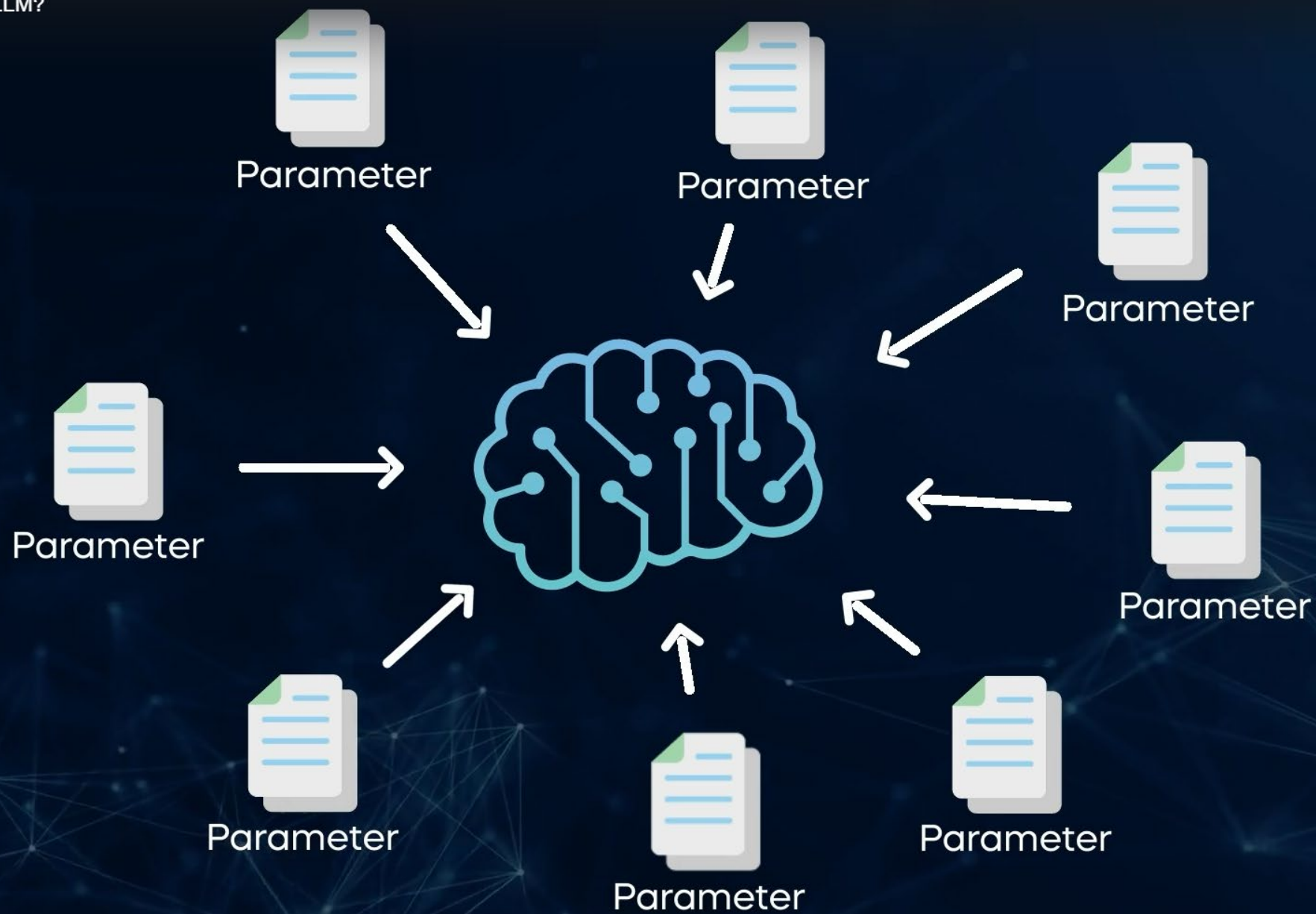
**Main features to large language models
which make them unique:**

1. The LLMs are larger compared to other models.
2. General purpose
3. Ability to be pre-trained and fine tuned.

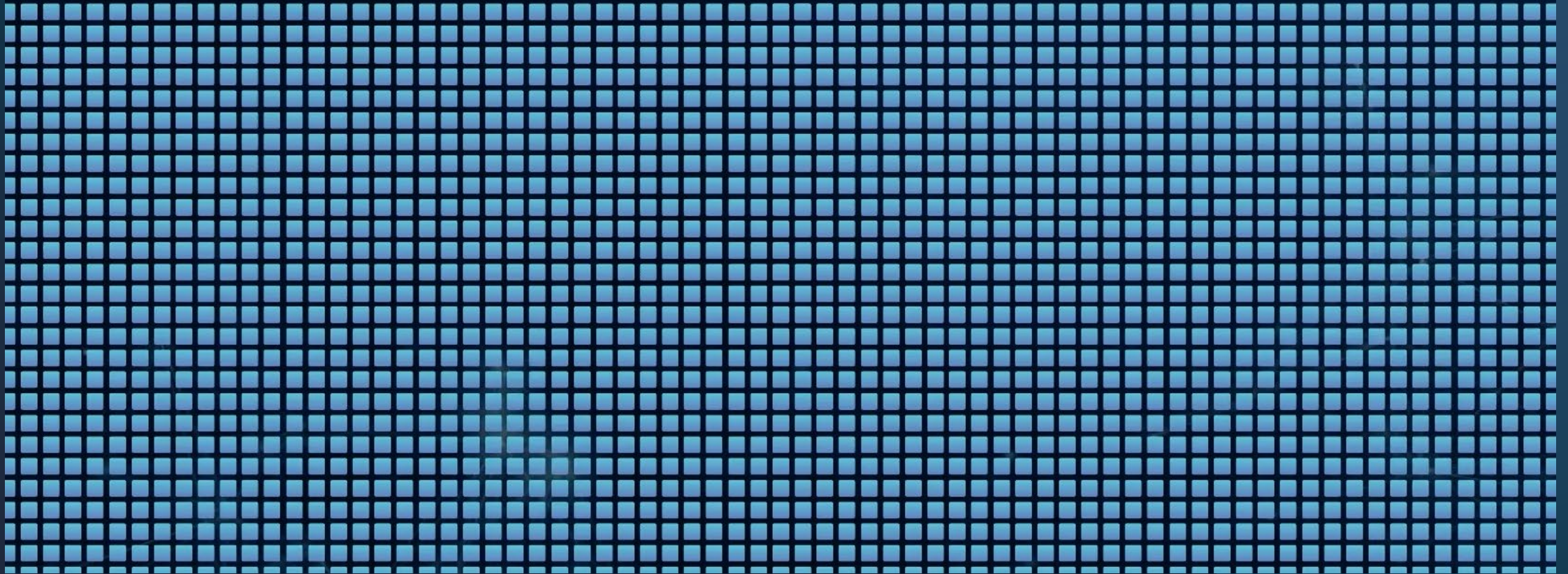


LLMs are characterized by their size

How large is an LLM?



**The more parameters a model has,
the better it can understand
and work with language**



15,000x increase in 5 years

Model size

GPT4
1.76T

GPT-3 1T
1 trillion

Megatron-Tuning
530B

GPT-3
175B

Transformers
65M

BERT
345M

GPT-2
1.5B

GPT-2 8B
8.3B

T5
11B

Turing-NLG
17B

MID

2018

2019

MID

LATE

2020

MID

LATE

2022



Text data includes things like:



Books:

on all sorts of topics, from adventure stories to science textbooks



Websites:

like news articles, blogs, and social media posts



Wikipedia:

They study articles to learn about facts, history, and general knowledge



Conversations:

They look at how people chat with each other online



Others:

recipes, movie reviews, scientific papers, etc.



General Purpose LLMs

“General purpose”



Model has been trained on a wide variety of text data from the Internet



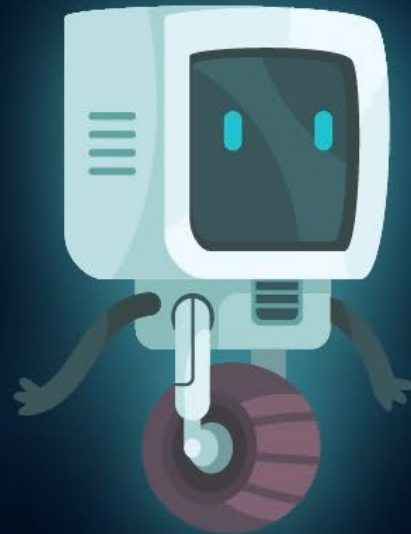
Designed to understand and generate human language in a broad and versatile manner

Multi-talented language tool that can assist with many different jobs



LLMs goal

Give general understanding of knowledge and how language works so it can be applied to more specific tasks later on



- Can understand language
- Can solve general problems

LLMs goal

Give general understanding of knowledge and how language works so it can be applied to more specific tasks later on



Fine-tune

**? What do we mean by
pre-trained and fine tuning a LLM**

Pre-Training and Fine-Tuning

SCHOOL SUBJECTS





Data Science

Math

Coding

Translation

LLM



Copy writing

Advices

Finance

Transport



Medical dataset



**Answering medical
questions**

Few-shot: where we use minimal data to train a model

Zero-shot: where the model can recognize things that it has not been explicitly taught in the training



I can do industry sepcific tasks
without specific data!

What LLMs are used For?





Content creation

- Write articles
- Blog posts
- Creative stories



Translation



Answering questions

- Answer questions on various topics
- Solve math problems



Chatbots

- Used to create chatbots and virtual assistants
- Can hold conversations with users



Sentiment analysis

- Analyze the sentiment in text



Summarization

- Summarize long articles or documents



Content recommendations

- Power recommendation systems



Generating code

- Generate code snippets
- Debug
- Explaining complex programming concepts



Medical diagnosis

- Analyzing medical records
- Suggesting possible diagnoses
- Staying updated with the latest medical research



Legal document reviews

- help identify relevant information
- help identify relevant information



Personalized marketing

- create personalized marketing campaigns
- recommending products or services

The Problems with Deep Learning Based Recurrent Neural networks (RNNs)

+ •



Different **Neural Networks**
deal with different problems

Recurrent Neural Network

English

I

am

going

to

buy

a

car

French

Je

vais

acheter

une

voiture

(The RNN holds this
word in context as it
waits for the next word.)

(The RNN holds this
word in context as it
waits for the next word.)

Different **Neural Networks**
deal with different problems

Recurrent Neural Network

✕ struggle when the input text is long



“The New York Times is a daily newspaper. It was first issued in 1851”

“The New York Times is a daily newspaper.



It was first issued in 1851



- We will not be able to truly understand the meaning of the text.
- It would take too long
- It won't be able to handle the size of the training data today

“The New York Times is a daily newspaper. It was first issued in 1851”



RRNs

“The New York Times is a daily newspaper.

It...



Transformers

“The New York Times is a daily newspaper.

It was first issued in 1851

The background is black with several abstract geometric elements. In the top left, there is a light orange semi-circle and two white zigzag lines. In the bottom left, there is a light green circle. In the bottom right, there is a light green semi-circle. A large black rectangle with a white border and a thin orange double-line border is centered on the page.

ATTENTION IS ALL YOU
NEED

“Attention is all you need”

Transformers + Attention mechanism

The attention mechanism enables a model to weigh the importance of different words or tokens in an input sequence when producing an output.

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 13].

^{*}Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†]Work performed while at Google Brain.

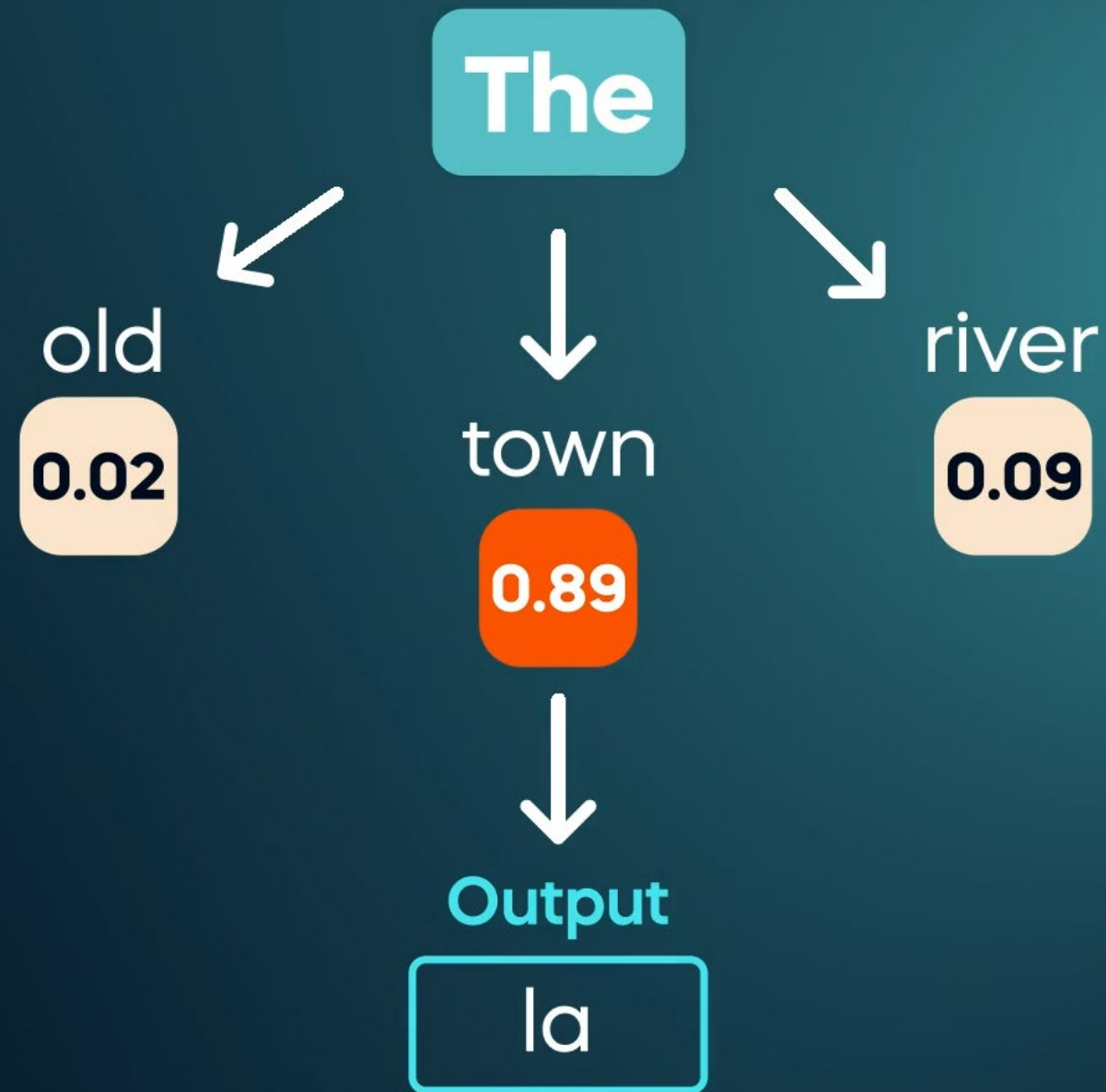
[‡]Work performed while at Google Research.

Input

Yest^{0.09}day, I visited ^{0.02}the^{0.89}old town near
the river. The streets were lined with
colorful houses, each telling a story of
its own. In the market, vendors were
selling fresh fruits and handcrafted
goods. Children played in the square,
their laughter mixing with the sounds
of the bustling town.

Output

Hier, j'ai visité



Simple translation model

French: “L’accord sur la zone économique européenne a été signé en août 1992”



English: The agreement on the area economic European was signed in August 1992

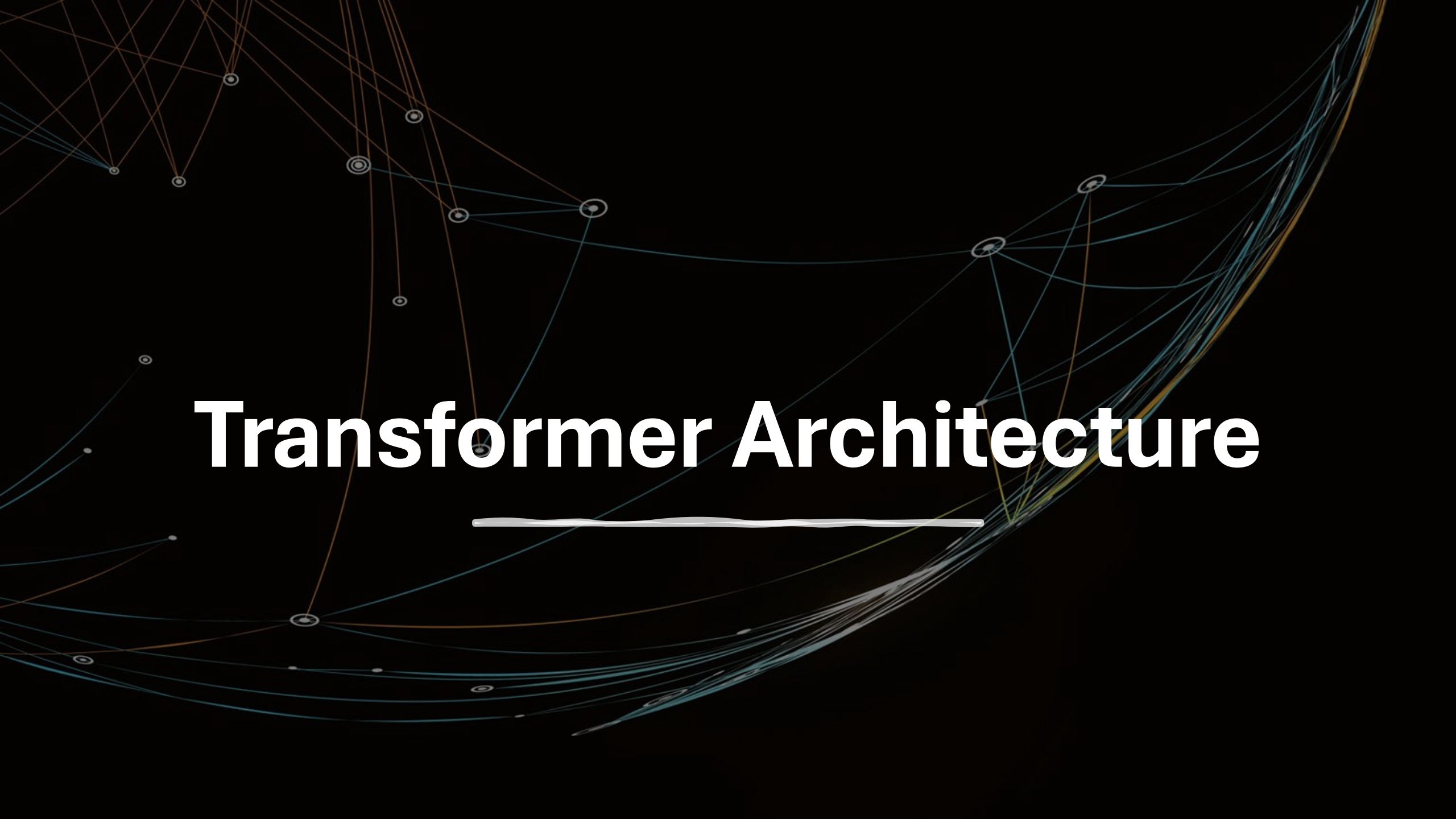
Simple translation model

French: “L'accord sur la zone économique européenne a été signé en août 1992”



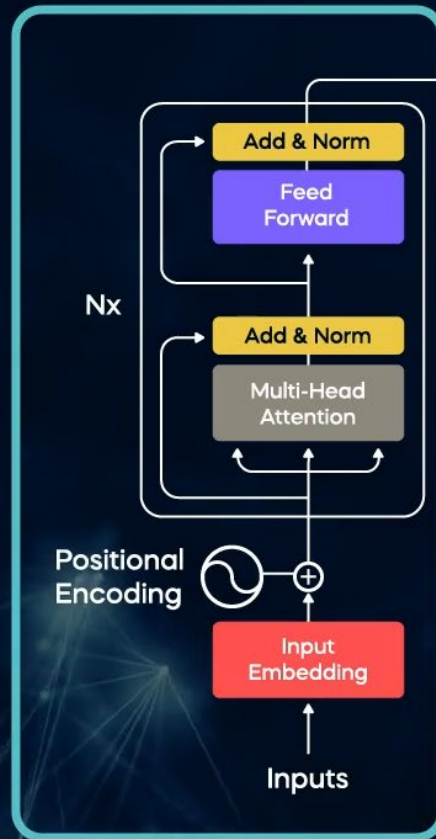
English: The agreement on the area economic European was signed in August 1992

European economic area

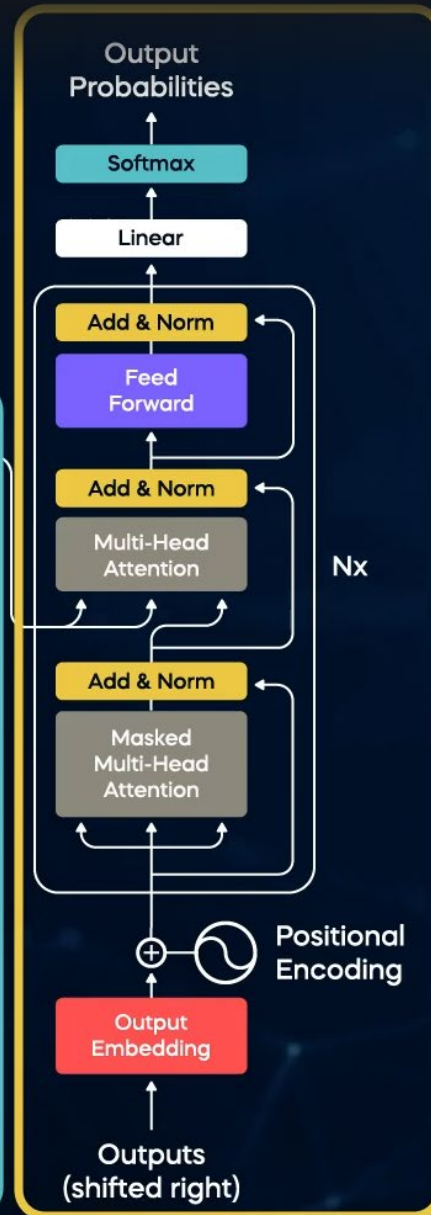


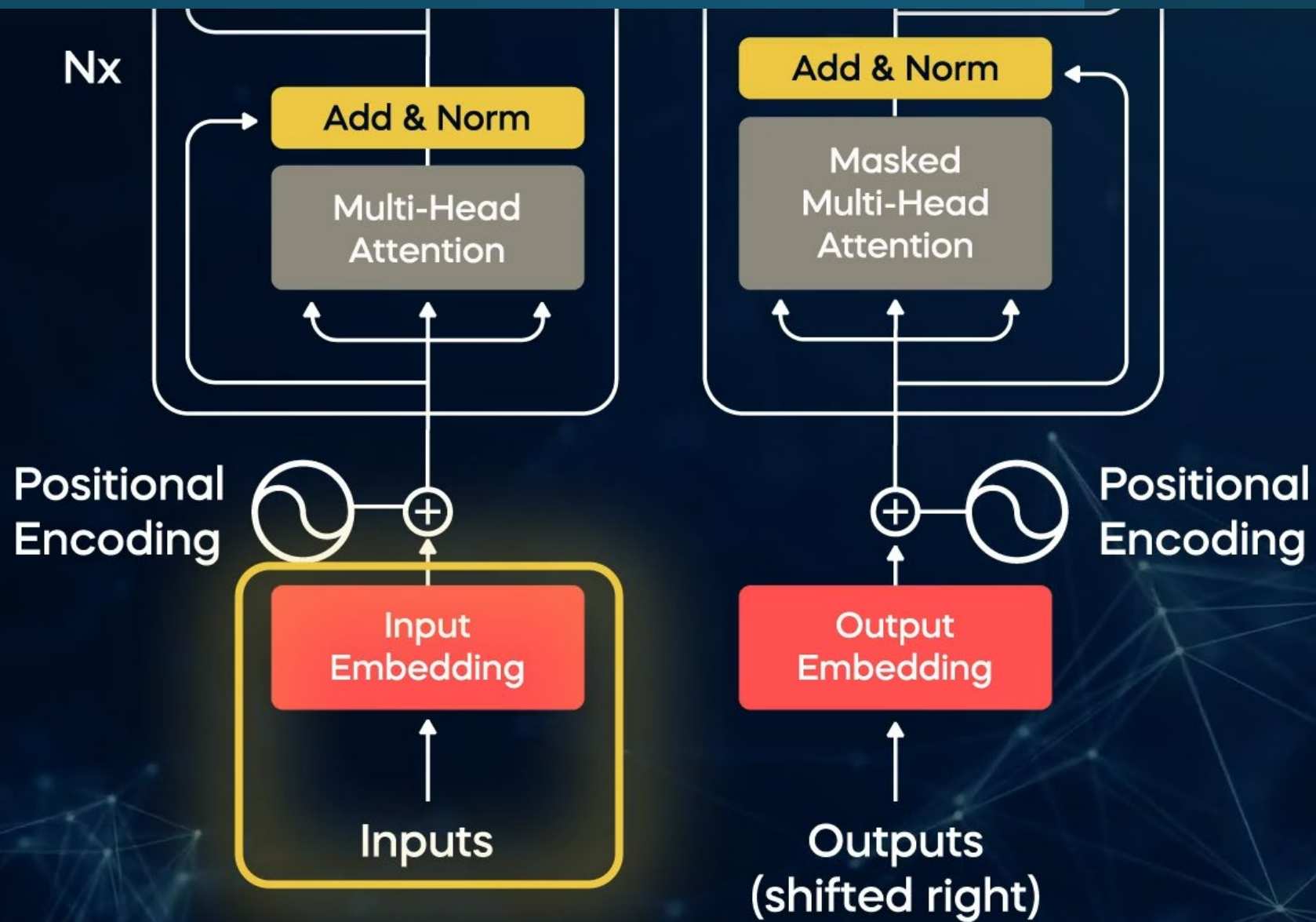
Transformer Architecture

Encoder



Decoder





Creating the input embeddings

1. Break down the input into tokens

Tokens = Words, subwords
or characters

"I love natural language processing"



["I", "love", "natural", "language", "processing"]

Creating the input embeddings

"I love natural language processing"



["I", "love", "natural", "language", "processing"]

"I" - 34

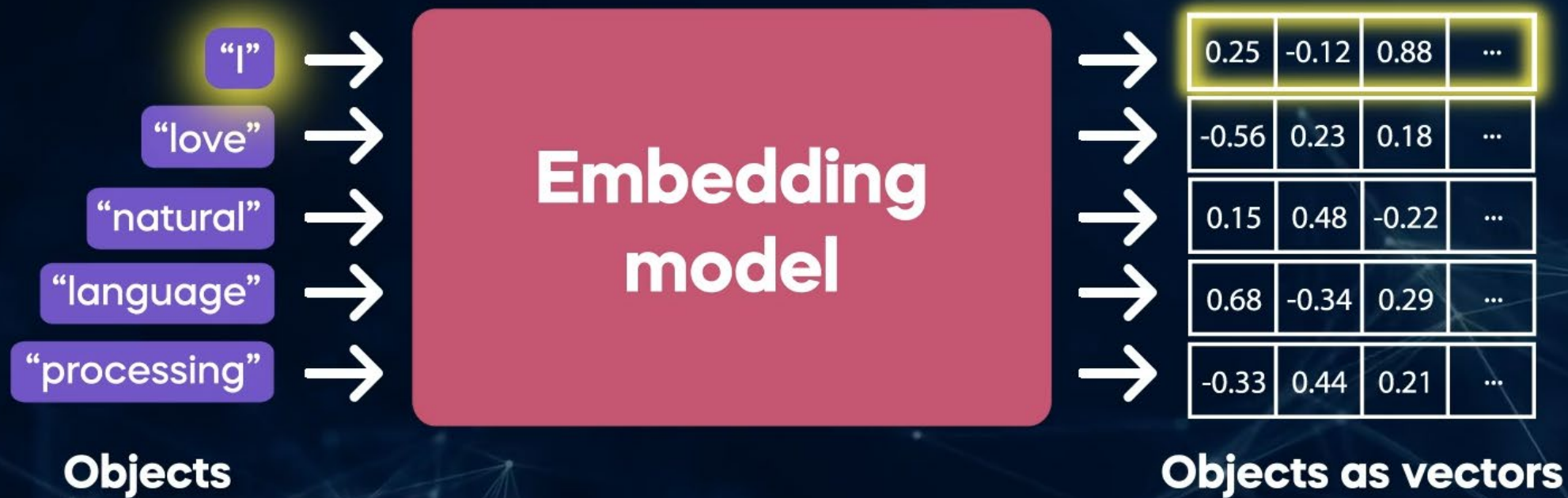
"love" - 56

"natural" - 96

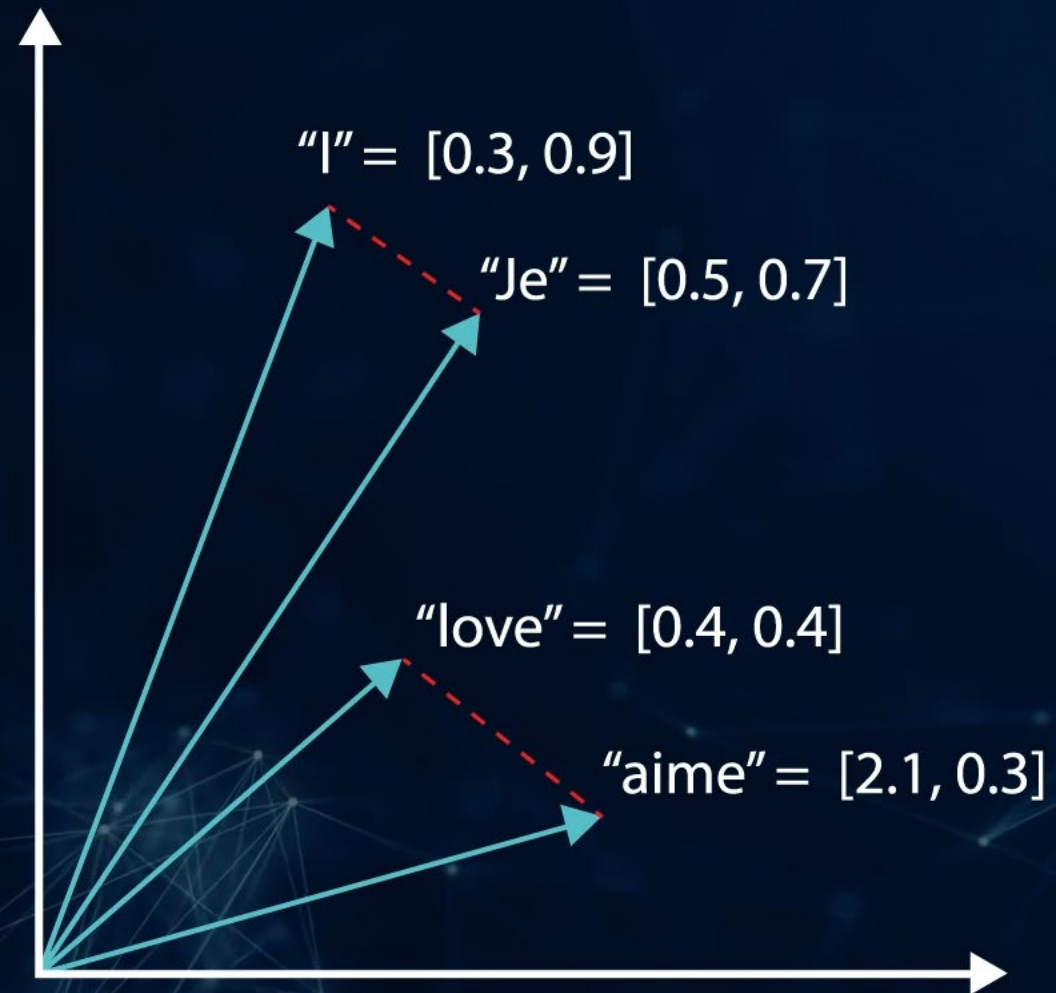
"language" - 12

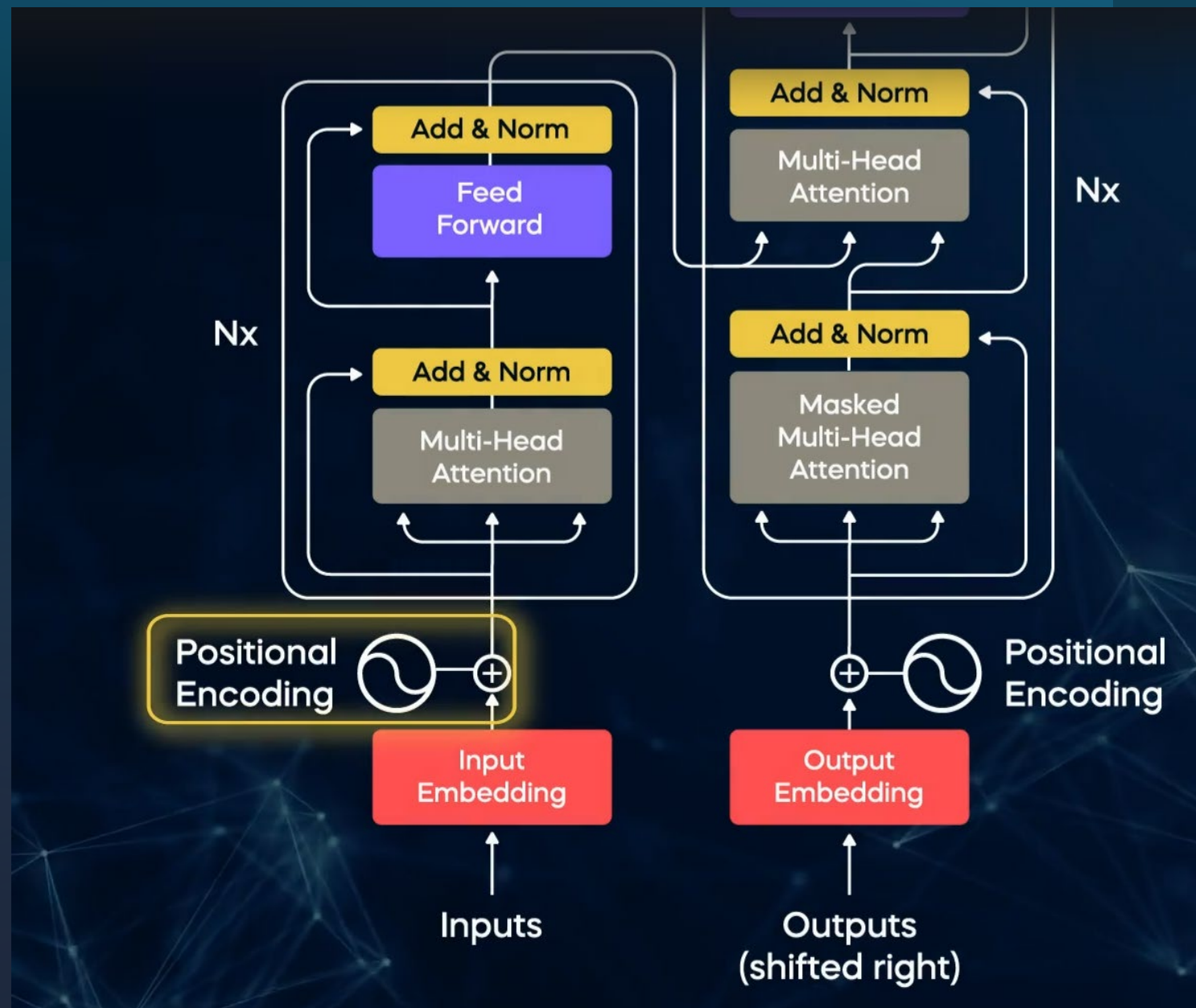
"processing" - 24

Creating the input embeddings



Creating the input embeddings





Positional Encoding

Input sequence I love natural language processing

Positional encoding

0

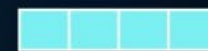
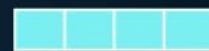
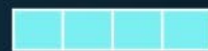
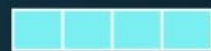
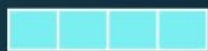
1

2

3

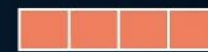
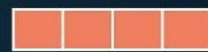
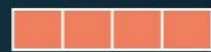
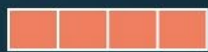
4

Positional encoding vector



+

Input embeddings



Encoder block

Padding

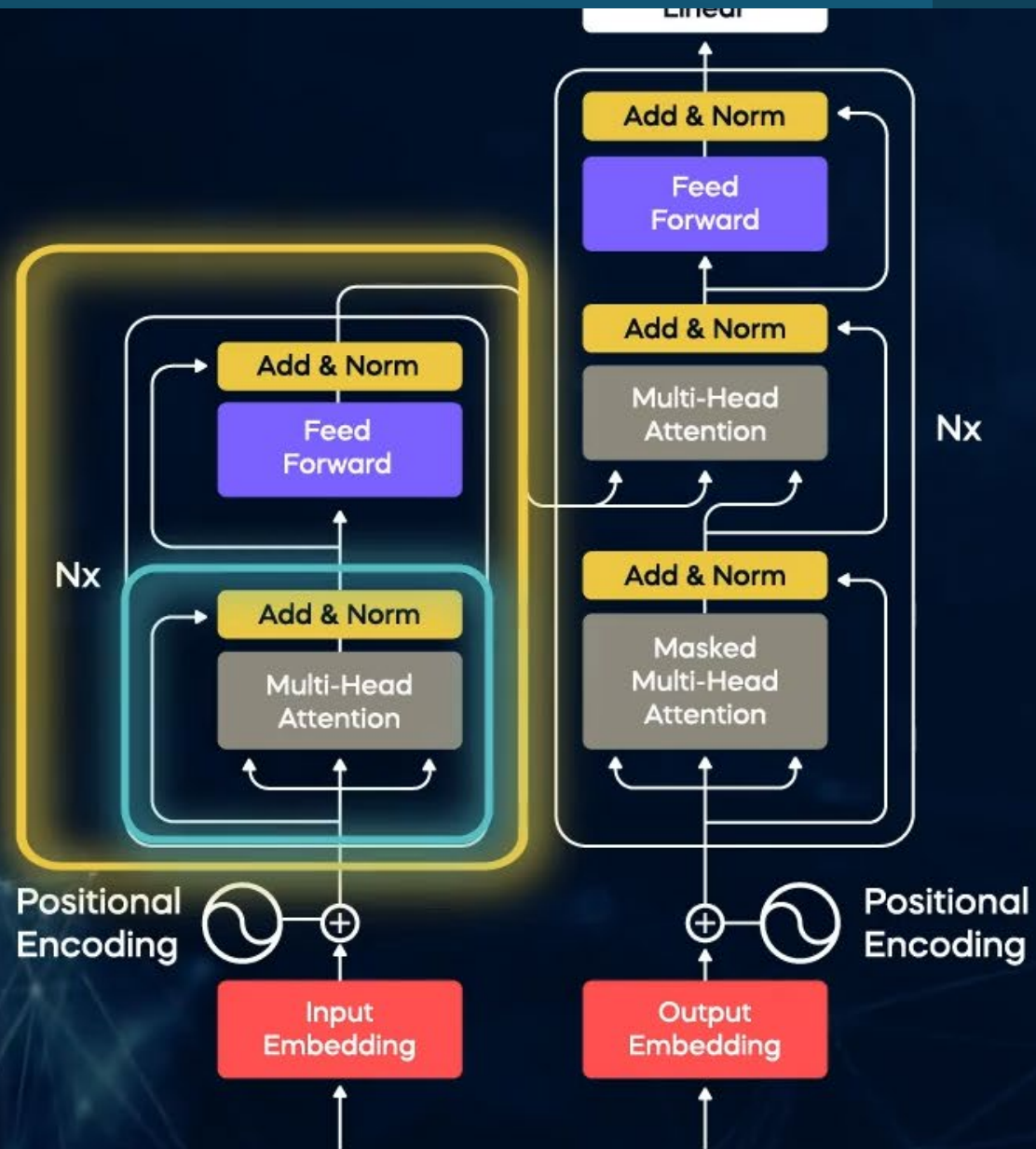
Adds special tokens or zeros to the embeddings of shorter sequences

Truncation

Removes tokens from longer sequences

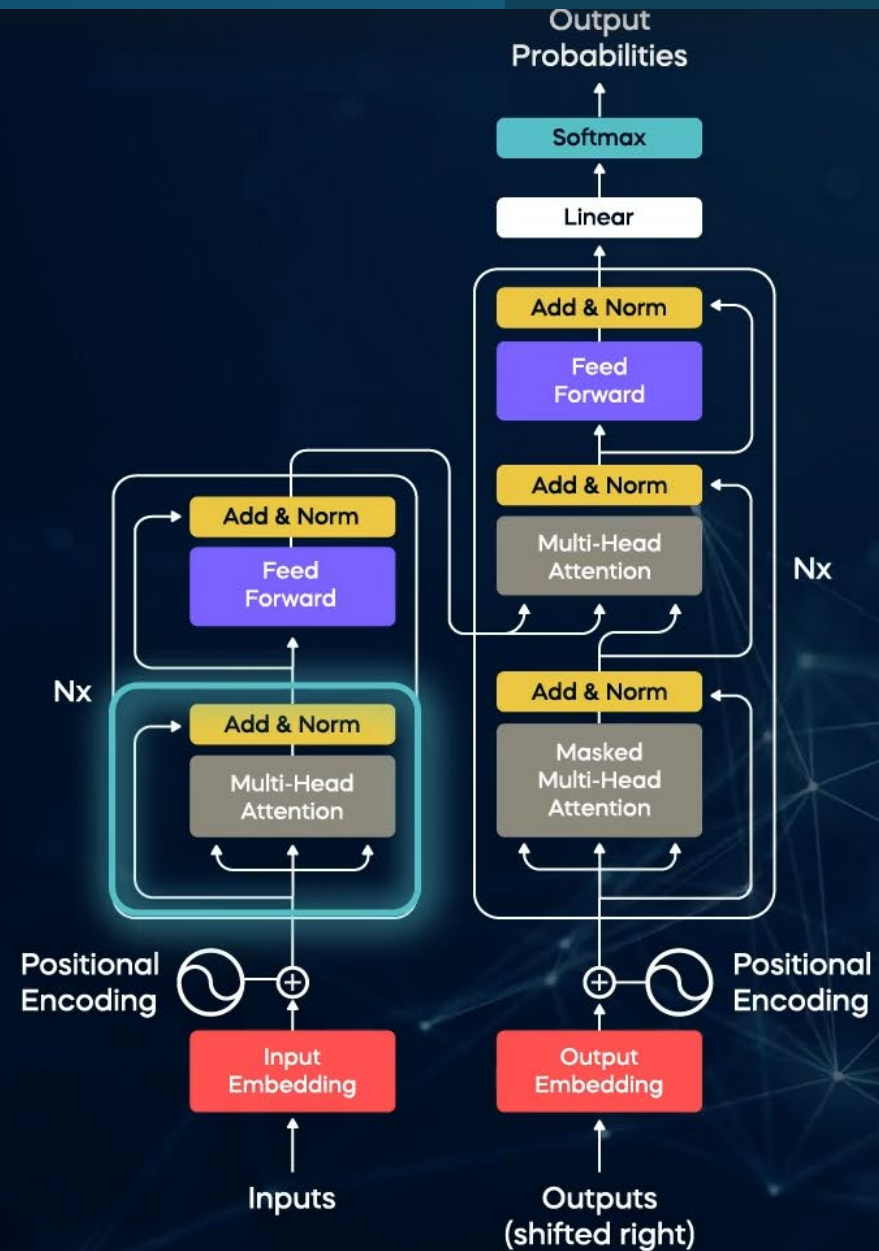
Encoder Block

Multi-Head Attention Layer



Multi-Head Attention Layer

- It allows the model to weigh the importance of different tokens
- Gives us attention vectors that capture contextual information between every token in the text



For each token in the input sequence, create three vectors:

Query Vector

This query vector represents the token's own "question" about the other tokens in the sequence

Input sequence:

A robot must obey the orders given to **it**



What is 'it' referring to?

For each token in the input sequence, create three vectors:

Key Vectors

- Hold information about all the other tokens in the sequence.
- Each key vector corresponds to a different token and contains information.
- Help determine how much attention should be given to that token

For each token in the input sequence, create three vectors:

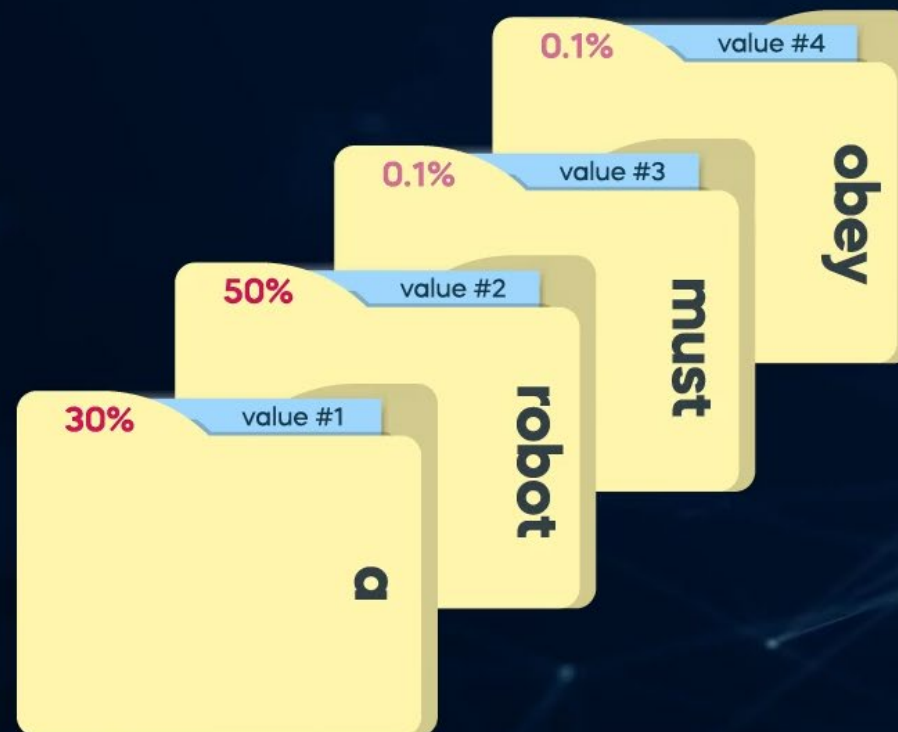
Value Vectors

- Associated with each token in the input sequence
- “Information containers”

For each token in the input sequence, create three vectors:

Value Vectors

Query vector

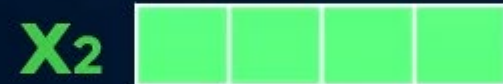
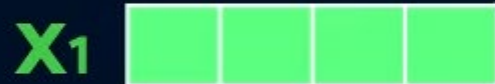


Input

robot

it

Embedding



Queries



Keys



Values

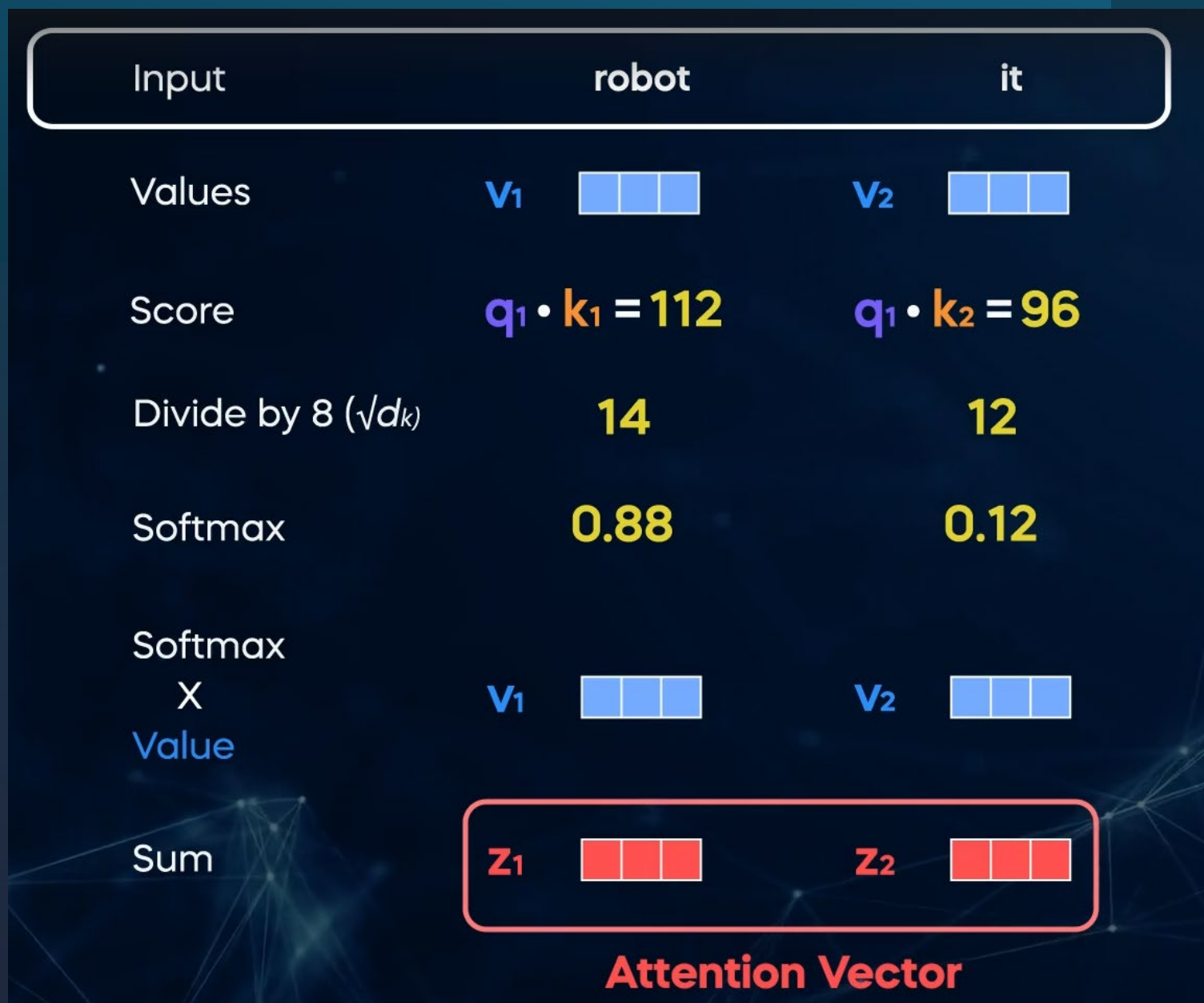


Score

$$q_1 \cdot k_1 = 112$$

$$q_1 \cdot k_2 = 96$$







Calculating attention separately on
eight different attention heads

**ATTENTION
HEAD #0**



**ATTENTION
HEAD #1**



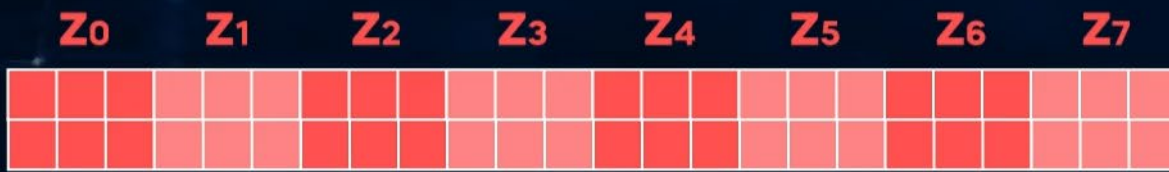
...

**ATTENTION
HEAD #7**

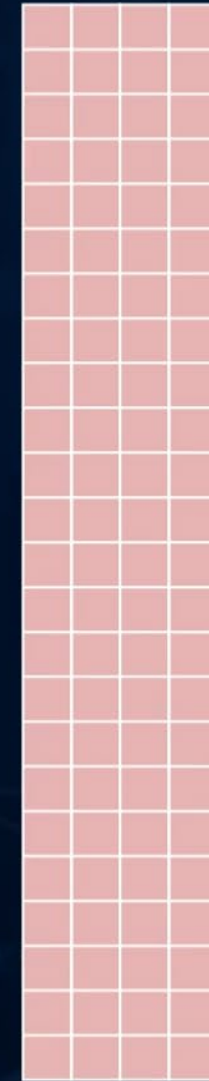


2) Multiply with a weight matrix W^o that was trained jointly with the model

1) Concatenate all attention heads



$\times$



W^o

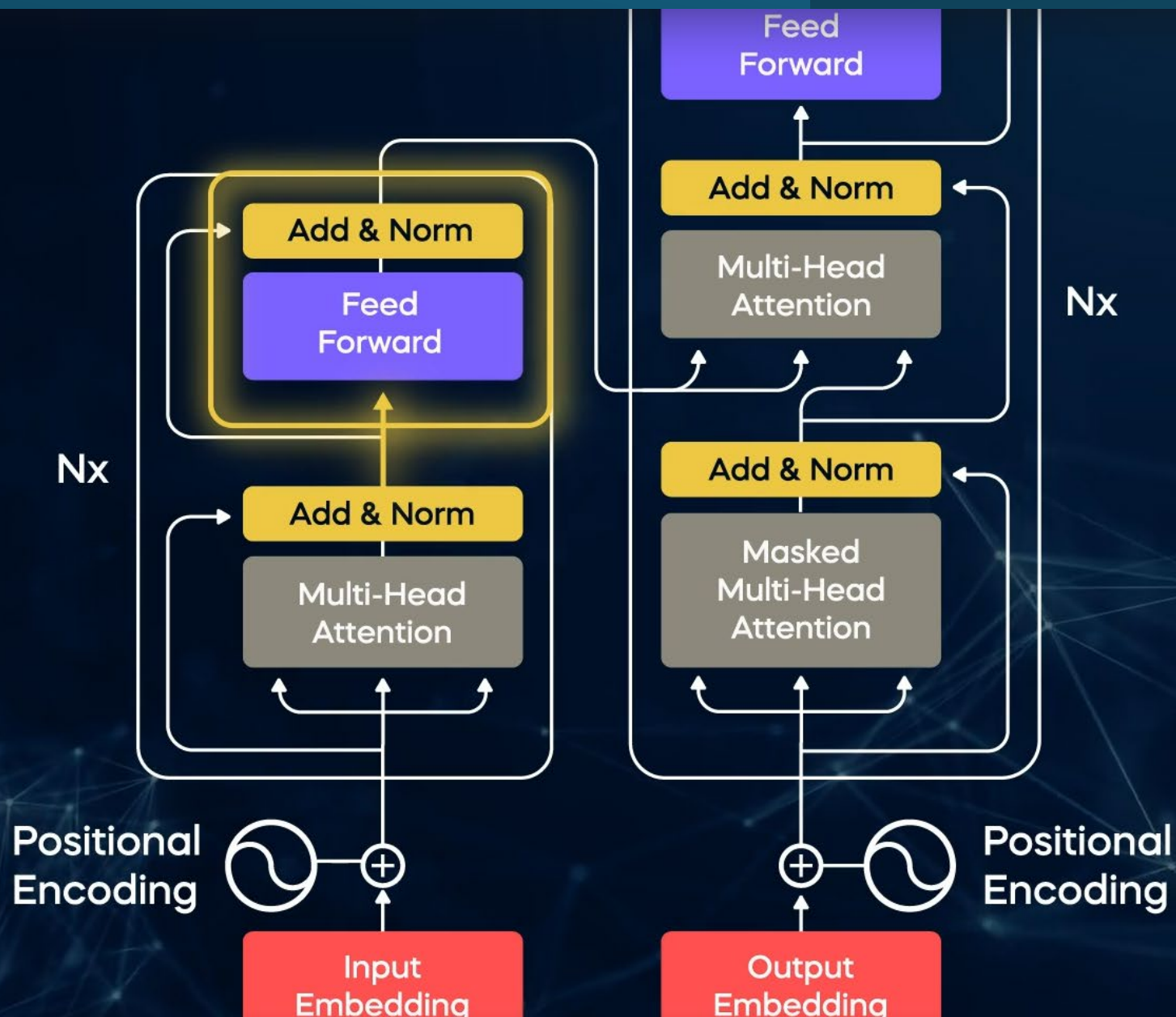
$=$

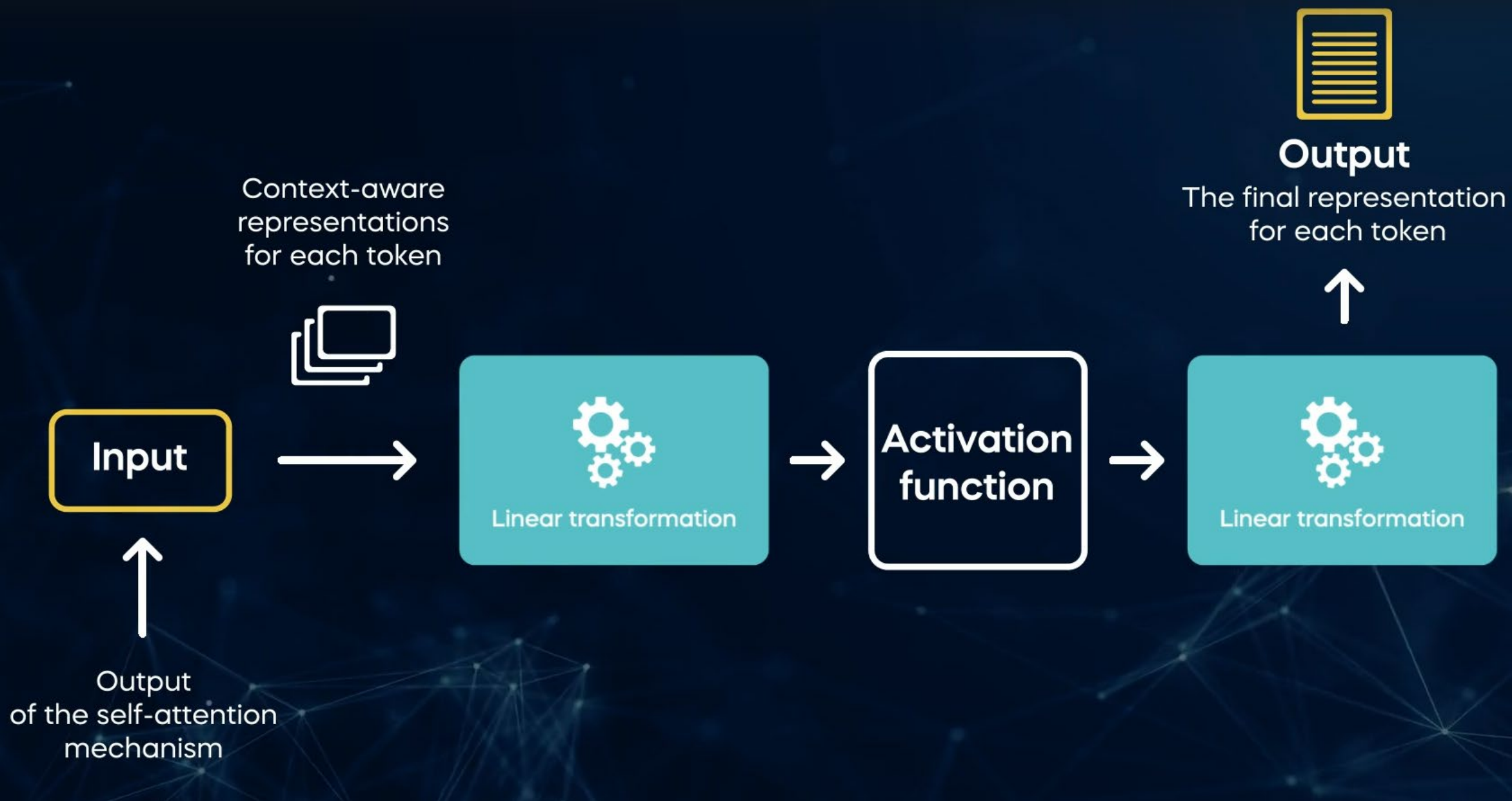


3) The result would be the Z matrix that captures information heads

Feed Forward Layer

- Models complex, relationships within the input sequence
- Processes and transforms the information captured during the self-attention mechanism



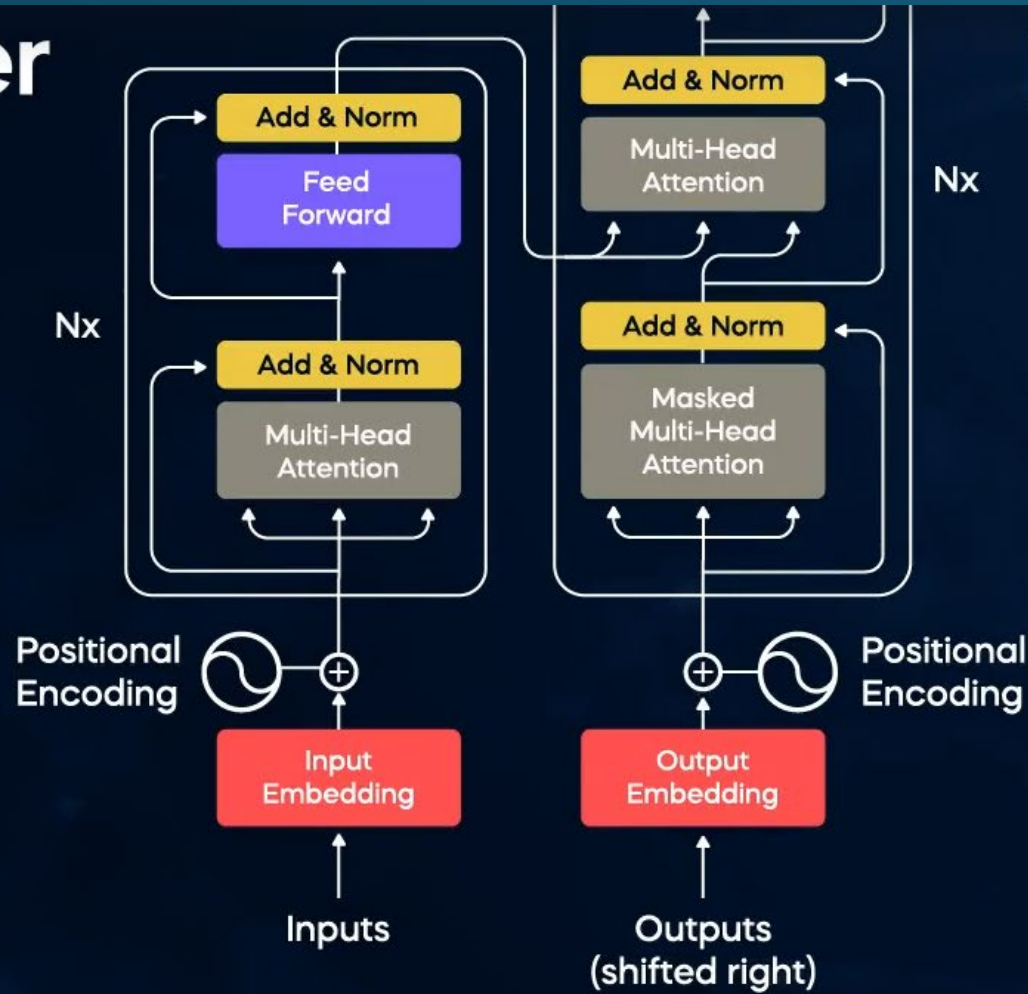


Feedforward layer:

- Acts as a set of neural network operations applied independently to each token's representation
- Helps the Transformer model capture and encode complex, non-linear relationships between tokens in the input sequence
- Applied independently
- It can be run in parallel



Encoder

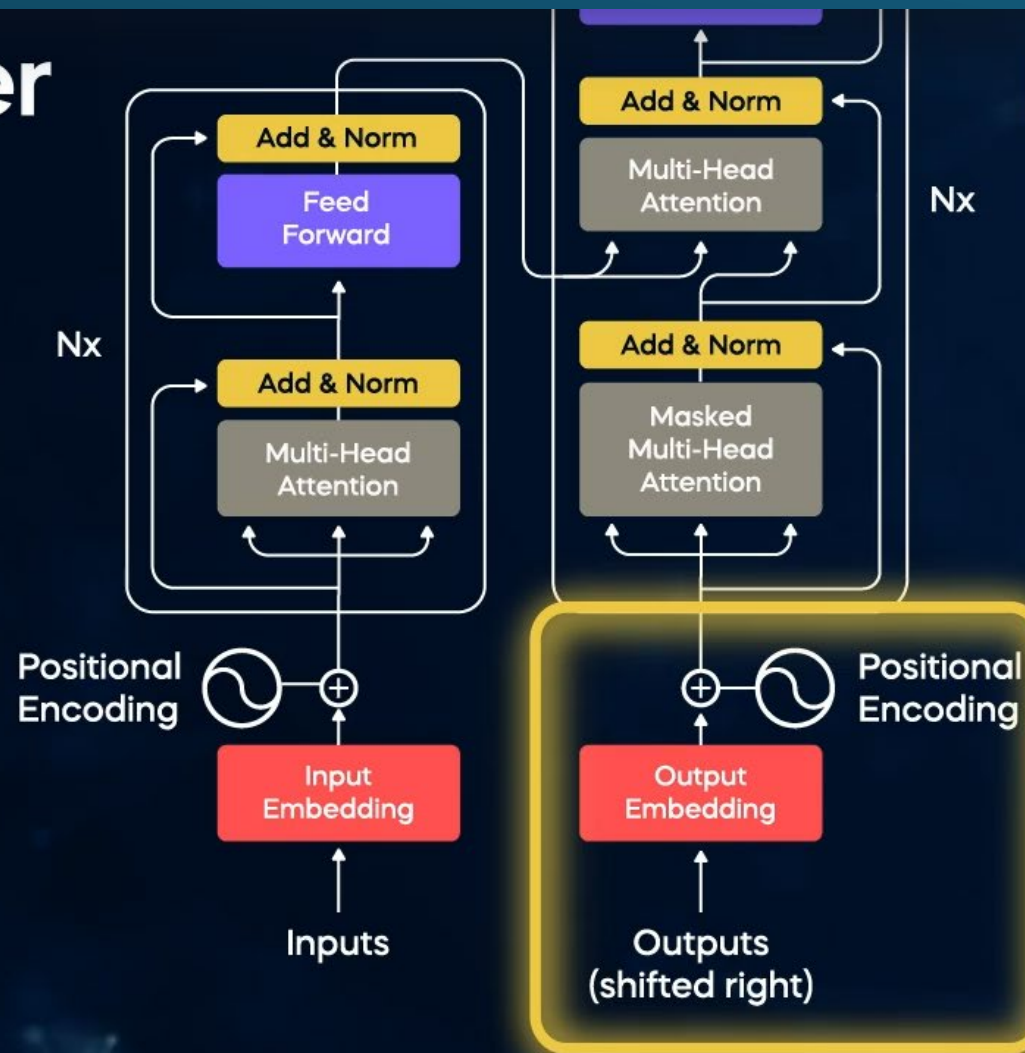


Decoder

French words

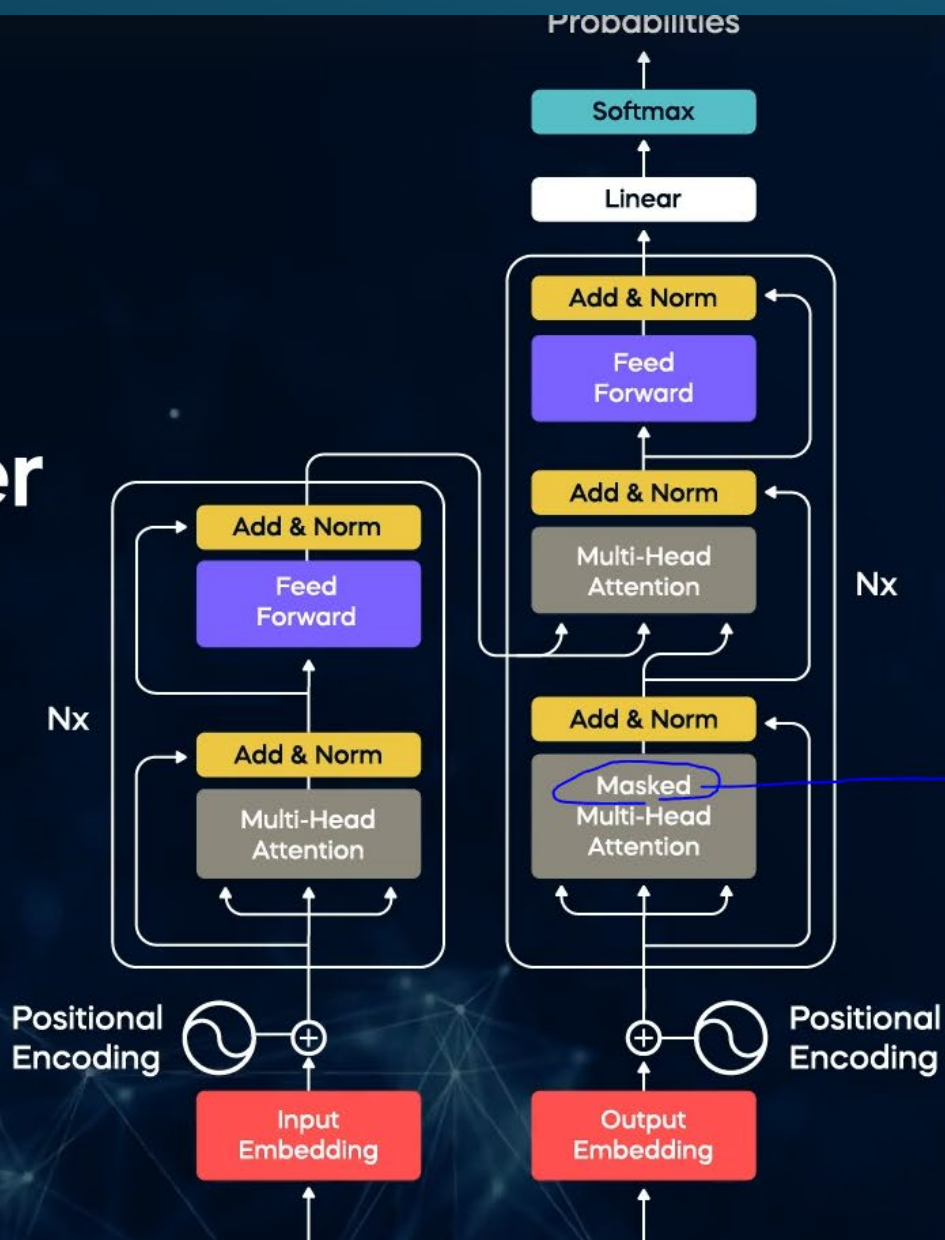
English words

Encoder



Decoder

Encoder



Decoder

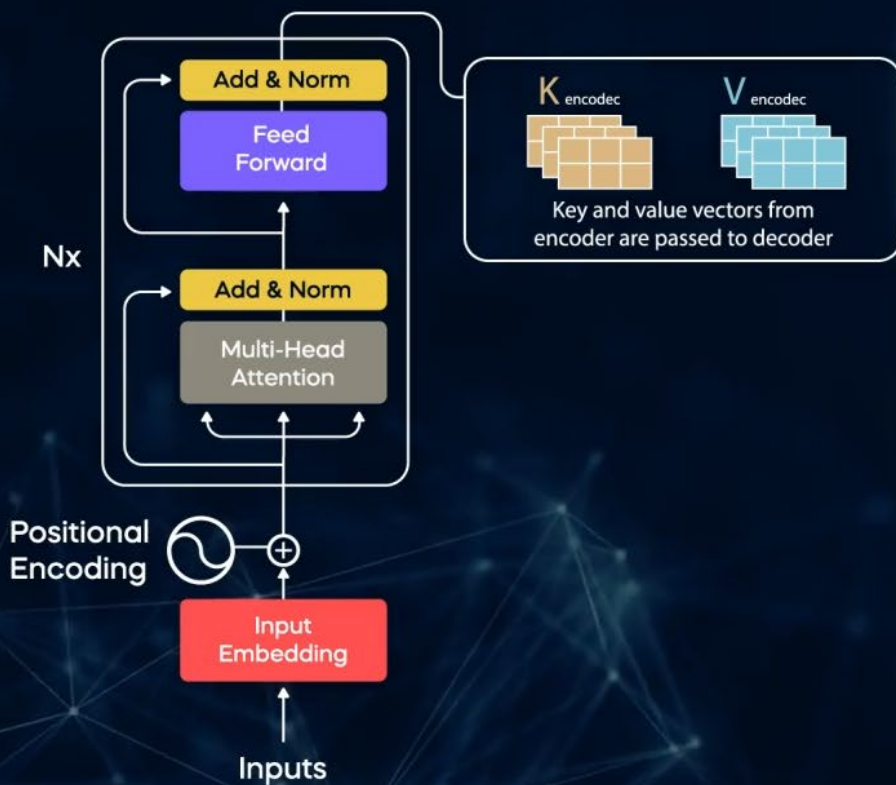
French words
(Attention vectors)

[0.32, 2.44, 7.43, ...]

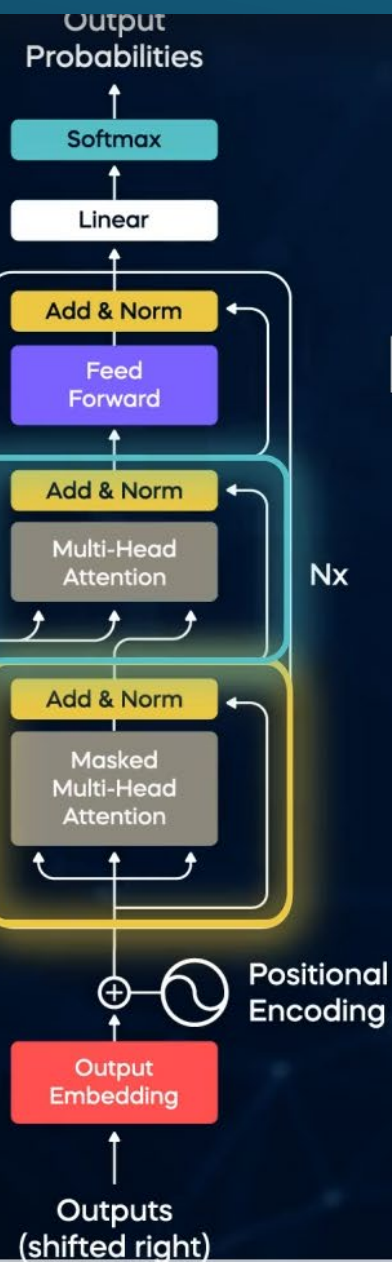


I love natural..

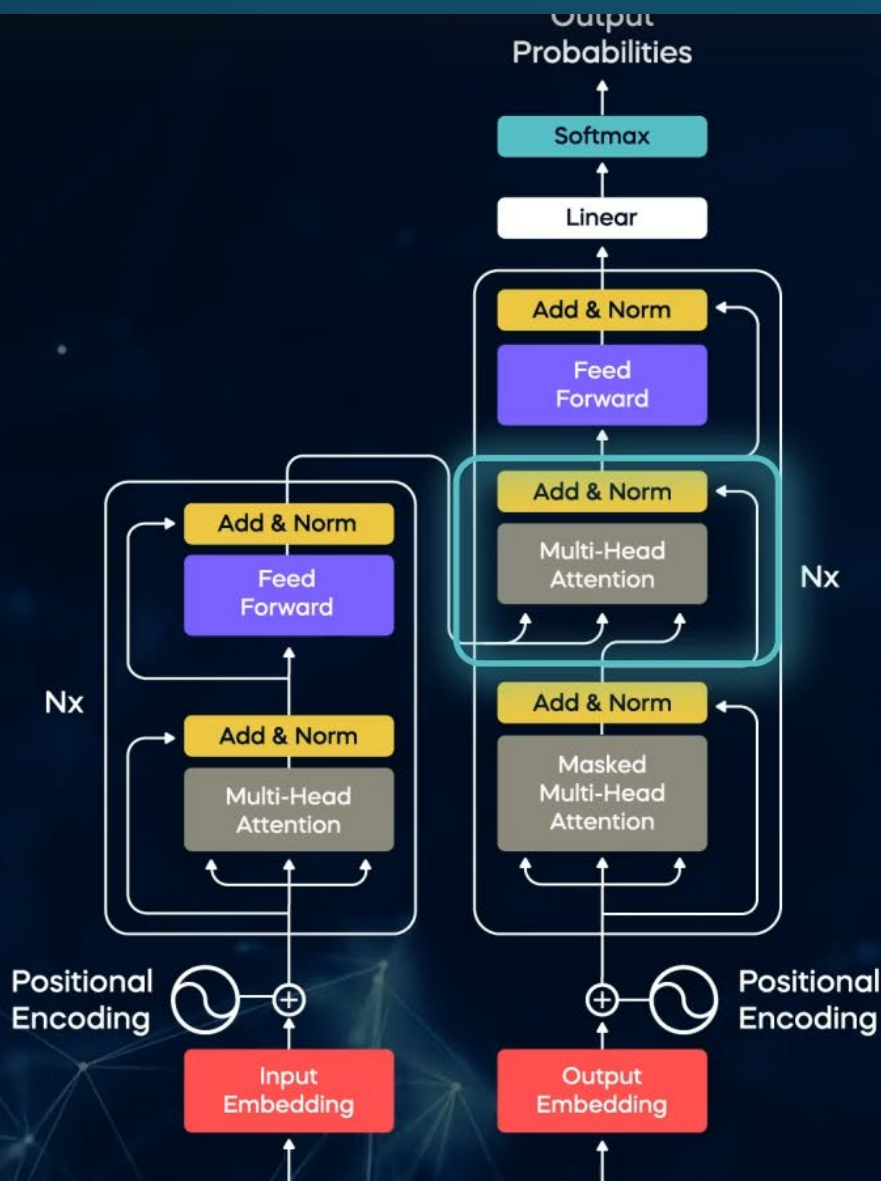
Encoder



Decoder



Encoder



Decoder

Multi-Head Attention

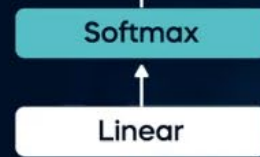
- Calculates attention scores between the current output token and the encoder
- Helps the model determine which parts of the input sequence are relevant when generating the next token

Encoder



Transforms the output into a probability distribution

Output Probabilities



Add & Norm

Feed Forward

Context Vector

Add & Norm

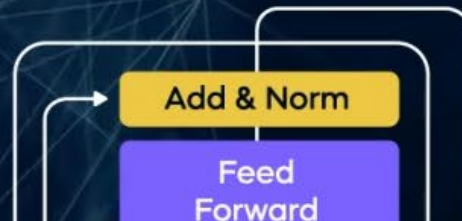
Multi-Head Attention

Manipulate the output into a useful format

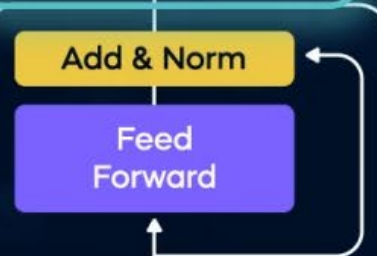
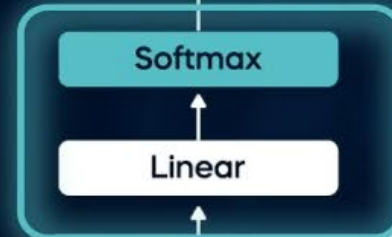
Decoder

Nx

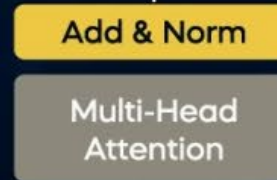
Encoder



Output
Probabilities



Context
Vector



Nx

Decoder

ChatGPT



Chat GPT



write poetry



write essays



create social media content



write code



GPT

Generative **P**re-trained **T**ransformer

Generative

- Can generate human like text
- Can continue the text in a coherent and contextually appropriate manner
- Very good at tasks like:



Text completion



Dialogue generation

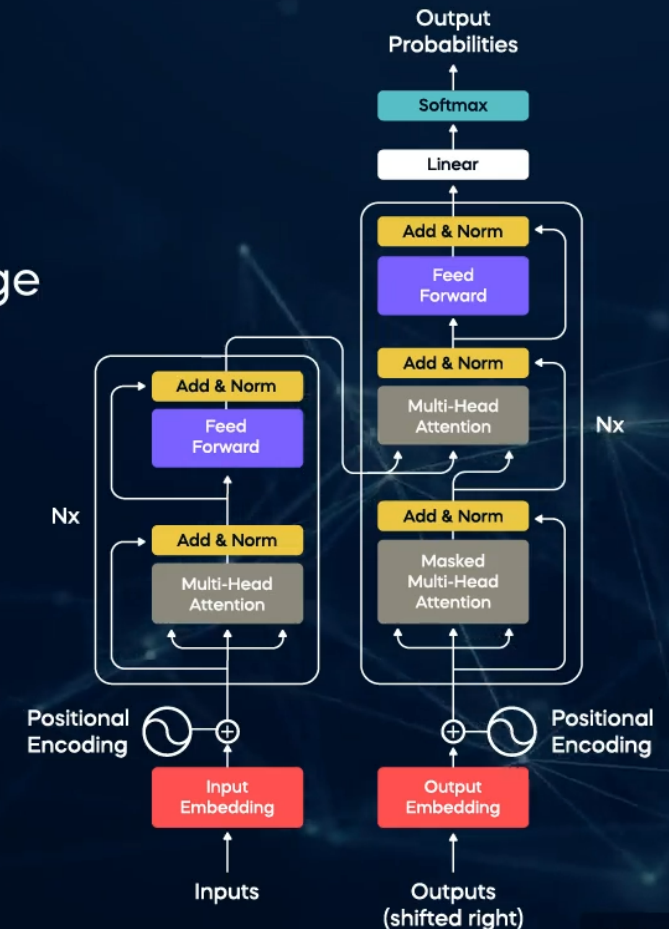


Content creation

Pre-trained

- learns from massive amount of text data
- Acquires deep understanding of language and grammar, and a broad vocabulary

Transformer





GPT-1



GPT-2



GPT-3



GPT-3.5



GPT-4



In the context of AI/ML and Data Science, Lambda refers to AWS Lambda, a serverless computing platform introduced by Amazon Web Services (AWS) in 2014. AWS Lambda was designed to simplify the deployment and management of code in the cloud.

Those seeking the application of NLP want an algorithm to comprehend texts for various purposes -- from identification and classification to entity extraction into business processes. All the benchmarks -- GLUE, SQuAD and RACE -- address this setup. In practical terms, this usually means extracting meaningful information out of unlabeled and semi-structured data in documents and related messages, such as emails.



GPT-1

- Released: 2018
- 117 million parameters
- Trained using unsupervised learning
- Gave state of the art performance for LAMBDA
- Gave Competitive performance for GLUE and SQuAD

GPT-2

- Released: 2019
- 1.5 billion parameters
- Trained on larger dataset
- Additional improvements like including a modified training objective, and a more efficient sampling algorithm for generating text

GPT-3

- Released: 2020
- 175 billion parameters
- Trained on larger dataset
- Additional improvements like better training methodology (Gshard) and Few shot learning Capability

GPT-3.5

- Released: 2022
- Three variants, 1.3 B, 6B and 175B parameters
- The main element is to eliminate toxic outputs to certain extent using RLHF (Reinforcement learning with human feedback)

GPT-4

- Released: 2023
- 1 trillion parameters
- It can take text and image as inputs and produces text outputs
- RLHF is used in the training

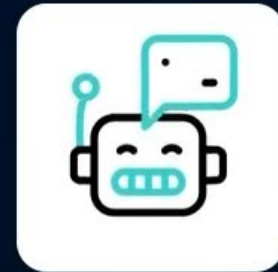


ChatGPT

For conversational interactions



Optimized for more natural
and engaging responses



Fine tuned for chatbots
and virtual assistants

Open AI is now
paid. Atleast 5\$ is
needed along with
a wait for 10-15
minutes for the
credits to work



Hi there,

We've updated the billing system for your [OpenAI API account](#). Instead of receiving a bill at the end of the month, you'll now need to pre-purchase credits to use the API. You can add credits to your account by visiting the [billing page](#). To learn more about prepaid billing, please see this [help center article](#).

No action is required from you at this time.

Please note that this change only applies to your OpenAI API account and Playground. It does not affect ChatGPT Plus subscriptions.

Best,
The OpenAI team

Your profile

Your details

Security

User API keys

Legacy



User API keys have been replaced by project API keys.

We recommend using project based API keys for more granular control over your resources. [Learn more](#)

[View project API keys](#)

Do not share your API key with others, or expose it in the browser or other client-side code. In order to protect the security of your account, OpenAI may also automatically disable any API key that has leaked publicly.

Enable tracking to see usage per API key on the [Usage page](#).

NAME	SECRET KEY	TRACKING ⓘ	LAST USED ⓘ	PERMISSIONS	
LLMsChinmaya	sk-...B1JN	Enabled	Never	All	 

[+ Create new secret key](#)

Default organization

If you belong to multiple organizations, this setting controls which organization is used by default when making requests with the API keys above.

Personal



Note: You can also specify which organization to use for each API request. See [Authentication](#) to learn more.

Billing

Overview

Payment methods

Billing history

Preferences

Pay as you go

Credit balance ⓘ

\$9.99

■ **Auto recharge is off**

When your credit balance reaches \$0, your API requests will stop working. Enable automatic recharge to automatically keep your credit balance topped up.

[Enable auto recharge](#)

Add to credit balance

Cancel plan



Payment methods

Add or change payment method



Billing history

View past and current invoices



Preferences

Manage billing information



Usage limits

Set monthly spend limits



Pricing

View pricing and FAQs

CTRL K**GET STARTED**

Overview

Quickstart

Concepts

Models

Libraries

Changelog

CAPABILITIES

Text generation

Vision

Function calling

JSON mode

Advanced usage

ENDPOINTS

Chat Completions

Overview

Getting started

Response Format

Fine-tuning

Batch

Image generation

Forum

Getting started

Chat models take a list of messages as input and return a model-generated message as output. Although the chat format is designed to make multi-turn conversations easy, it's just as useful for single-turn tasks without any conversation.

An example Chat Completions API call looks like the following:

```
python   
1 from openai import OpenAI  
2 client = OpenAI()  
3  
4 response = client.chat.completions.create(  
5     model="gpt-4o-mini",  
6     messages=[  
7         {"role": "system", "content": "You are a helpful assistant."},  
8         {"role": "user", "content": "Who won the world series in 2020?"},  
9         {"role": "assistant", "content": "The Los Angeles Dodgers won the World Series in 2020."},  
10        {"role": "user", "content": "Where was it played?"}  
11    ]  
12 )
```

To learn more, you can view the full [API reference documentation](#) for the Chat API.

Including conversation history is important when user instructions refer to prior messages. In the example above, the user's final question of "Where was it played?" only makes sense in the context of the prior messages about the World Series of 2020. Because the models have no memory of past requests, all relevant information must be supplied as part of the conversation history in each request. If a conversation cannot fit within the model's token limit, it will need to be [shortened](#) in some way.

 To mimic the effect seen in ChatGPT where the text is returned iteratively, set the [stream](#) parameter to true.

Chat Completions response format



System message:

You are a helpful tool that extracts the Keywords from a given block of text



User message:

"Transformers have revolutionized AI, particularly in natural language processing. As an AI professional, it's essential to understand these models. Mastering transformers is key to staying updated with AI advancements"



Assistant message:

"Transformers, AI, natural language processing, models, transformers"



System message:

Instructions to the model
How the model should respond and behave



User message:

User Input(block of text)



Assistant message:

Output to the user input
(the keywords extracted from the block of text)

Use Case	Temperature	Top_p	Description
Code Generation	0.2	0.1	Generates code that adheres to established patterns and conventions. Output is more deterministic and focused. Useful for generating syntactically correct code.
Creative Writing	0.7	0.8	Generates creative and diverse text for storytelling. Output is more exploratory and less constrained by patterns.
Chatbot Responses	0.5	0.5	Generates conversational responses that balance coherence and diversity. Output is more natural and engaging.
Code Comment Generation	0.3	0.2	Generates code comments that are more likely to be concise and relevant. Output is more deterministic and adheres to conventions.
Data Analysis Scripting	0.2	0.1	Generates data analysis scripts that are more likely to be correct and efficient. Output is more deterministic and focused.
Exploratory Code Writing	0.6	0.7	Generates code that explores alternative solutions and creative approaches. Output is less constrained by established patterns.



LangChain



GPT data can be outdated!





LangChain



+



GOAL:

A tool that helps you study



Application



Documents

Split into
chunks



Create
Embeddings





Data loaders



Text file loader



PDF loader



HTML loader